

QUINN JACKSON

Ottawa, Ontario

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EDUCATION

University of Ottawa

Sep. 2022 – Dec. 2026

Honours BSc in Computer Science

9.63/10 GPA

Honours Thesis (in progress): An empirical investigation into how fine-tuning, retrieval augmentation, and personalization affect the detectability and safety-relevant stylistic signatures of large language models.

EXPERIENCE

Public Safety Canada

AI Engineer

Sep. 2025 – Dec. 2025

- Designed data ingestion pipelines that continuously ingested and normalized high-volume public social media, OSINT feeds, and internal government information systems to support time-sensitive emergency management.
- Architected multi-stage triage pipelines for rapidly evolving social media and OSINT ecosystems, enabling analysts to prioritize high-signal content amidst large-scale information volatility and coordinated misinformation activity.
- Implemented a verifiable analysis framework that combined AI-based content classification with provenance checks, reverse-image and cross-source verification, and structured human review, ensuring defensible intelligence products for senior decision-makers in emergency and national security contexts.

DistillerSR

Software Developer

Jan. 2025 – Apr. 2025

- Trained and integrated a question suitability classification model, enhancing NLP accuracy by categorizing generated prompts into specific error types, ensuring contextual appropriateness in highly regulated review environments.
- Developed responsive components and animations for the Smart Evidence Extraction feature, following company design standards and modern UI best practices to improve user clarity during AI-driven data extraction workflows.
- Implemented automated and customizable PubMed reference generation by integrating GROBID for structured citation extraction from PDFs, streamlining literature sourcing and reducing manual bibliography creation efforts.

SELECTED PROJECTS

Election Forecast | React, TypeScript, D3.js

- Developed a predictive statistical model based on Bayesian inference, forecasting Canadian federal election outcomes across 343 ridings by analyzing and correlating historical data, polling trends, and socio-political factors.
- Built an interactive front-end using React, TypeScript, and D3.js, featuring dynamic visualizations, real-time updates, and intuitive user controls to explore projected vote shares, seat distributions, and winning probabilities.
- Created advanced electoral analysis tools, including real-time maps, interactive graphs, and live-updated poll data scraped from multiple sources to visualize party performance, regional trends, and voter shifts.

Ottawa Digital Twin | Python, ArcGIS, TypeScript

- Cleaned, normalized, and spatially joined heterogeneous geospatial datasets, including zoning bylaws, proposed amendments, building footprints, and ward boundaries to produce unified, analysis-ready GeoJSON layers.
- Designed a modern, mobile-responsive, and accessible 3D web interface enabling stakeholders to explore proposed zoning changes, fostering transparency and community engagement while prioritizing usability for diverse audience.
- Presented at HUBBUB, earning Top Winner and People's Choice awards, delivered the tool to the City for councillor-led consultations on the 2026 zoning by-law update.

Open Source Search | Python, OpenCV, PyTorch

- Built a data pipeline that scraped and aggregated images and text from public social media and open web sources, normalizing metadata such as timestamps, user identifiers, URLs, and EXIF into an analysis-ready store.
- Designed a multi-stage detection system for both images and text, combining perceptual hashing, error level analysis, vanishing point and perspective checks, and AI-generated-text classifiers to flag likely synthetic or manipulated content across high-volume streams, developing interpretable forensic outputs that surface suspicious regions, geometric inconsistencies, and model scores for each item, later expanding on these paradigm at Public Safety Canada.

SKILLS

- Languages:** Python, Java, PHP, JavaScript, TypeScript, SQL, HTML, CSS, C, Go, Bash
- Frameworks:** React, Laravel, Angular, Spring Boot, D3.js, Node.js, Flask, FastAPI, Jest, JUnit, Jenkins, Cypress
- Developer Tools:** Git, Docker, Tableau, MySQL, ArcGIS, Postman, Jira, Firebase, Android Studio, Pandas, NumPy